LAB SHEET 2

OBJECTIVE

To understand and implement SQL transactions using COMMIT, SAVEPOINT, and ROLLBACK, retrieve records using SELECT queries, and perform INSERT, UPDATE, and ALTER operations on tables.

Table Structures:

CLASS

D (NUMBER) and Name (VARCHAR).

The initial records in the table are (1, 'Bravo'), (2, 'Alex'), and (4, 'Cheng').

EMPLOYEES

Employee_Id (NUMBER), First_Name (VARCHAR), Last_Name (VARCHAR), Email (VARCHAR), Phone_Number (VARCHAR), Hire_Date (DATE), Job_Id (VARCHAR), Salary (NUMBER), Commission_Pct (NUMBER), Manager_Id (NUMBER), and Department_Id (NUMBER).

DDL Questions (Data Definition Language):

1. How do you create a table named CLASS with appropriate data types?

   QUERY

```sql
CREATE DATABASE DBMS2_Parinita;
USE DBMS2_Parinita;
CREATE TABLE CLASS (
    ID numeric,
    NAME VARCHAR(20)
);
INSERT INTO CLASS VALUES (1, 'Bravo');
INSERT INTO CLASS VALUES (2, 'Alex');
INSERT INTO CLASS VALUES (4, 'Cheng');
COMMIT;
SELECT* FROM CLASS;
```

   OUTPUT

Result Grid

ID	NAME
1	Bravo
2	Alex
4	Cheng

2. How do you create a table named EMPLOYEES with the specified attributes?

QUERY

```sql
CREATE DATABASE DBMS2_Parinita;
USE DBMS2_Parinita;
CREATE TABLE EMPLOYEES (
    Employee_Id INT,
    First_Name VARCHAR(20),
    Last_Name VARCHAR(20),
    Email VARCHAR(30),
    Phone_Number VARCHAR(15),
    Hire_Date DATE,
    Job_Id VARCHAR(15),
    Salary DECIMAL(10,2),
    Commission_Pct DECIMAL(5,2),
    Manager_Id INT,
    Department_Id INT
);
```

OUTPUT

Field	Type	Null	Key	Default	Extra
Employee_Id	int(11)	YES		NULL	
First_Name	varchar(20)	YES		NULL	
Last_Name	varchar(20)	YES		NULL	
Email	varchar(30)	YES		NULL	
Phone_Number	varchar(15)	YES		NULL	
Hire_Date	date	YES		NULL	
Job_Id	varchar(15)	YES		NULL	
Salary	decimal(10,2)	YES		NULL	
Commission_Pct	decimal(5,2)	YES		NULL	
Manager_Id	int(11)	YES		NULL	
Department_Id	int(11)	YES		NULL	

3. How do you create a table named Client_master with the specified attributes?

QUERY

```sql
CREATE DATABASE DBMS2_Parinita;
USE DBMS2_Parinita;
CREATE TABLE Client_master (
    ClientNo VARCHAR(10),
    Name VARCHAR(20),
    Address VARCHAR(30),
    City VARCHAR(20),
    State VARCHAR(20),
    Pincode INT,
    Bal_Due DECIMAL(10,2)
);
DESC Client_master;
```

OUTPUT

Result Grid | Filter Rows: | Export: | Wra

Field	Type	Null	Key	Default	Extra
ClientNo	varchar(10)	YES		NULL	
Name	varchar(20)	YES		NULL	
Address	varchar(30)	YES		NULL	
City	varchar(20)	YES		NULL	
State	varchar(20)	YES		NULL	
Pincode	int(11)	YES		NULL	
Bal_Due	decimal(10,2)	YES		NULL	

4. What SQL statement would you use to rename the table Client_master to Client12?

QUERY

```
CREATE DATABASE DBMS2_Parinita;
USE DBMS2_Parinita;
RENAME TABLE Client_master TO Client12;
```

OUTPUT

Result Grid | Filter Rows

Tables_in_student
class
client12
employees

5. How do you display the bal_due column heading as "BALANCE"?

QUERY

```
CREATE DATABASE DBMS2_Parinita;
USE DBMS2_Parinita;
SELECT Bal_Due AS BALANCE
FROM Client12;
```

OUTPUT

Result Grid

BALANCE

DML QUESTION (DATA MANIPULATION LANGUAGE)

Part A: Transactions and Savepoints (CLASS Table)

Give the output of the following SQL script:

- **INSERT INTO class VALUES (5,'Rahul');**
- **COMMIT;**

QUERY

```
CREATE DATABASE DBMS2_Parinita;
USE DBMS2_Parinita;
INSERT INTO CLASS VALUES (5,'Rahul');
COMMIT;
SELECT* FROM CLASS;
```

OUTPUT

Result Grid

ID	NAME
1	Bravo
2	Alex
4	Cheng
5	Rahul

- **UPDATE class SET name = 'Abhijeet' WHERE id='5';**

QUERY

```
CREATE DATABASE DBMS2_Parinita;
USE DBMS2_Parinita;
UPDATE CLASS
SET Name = 'Abhijeet'
WHERE ID = 5;
```

OUTPUT

Result Grid Filter

ID	Name
1	Bravo
2	Alex
4	Cheng
5	Abhijeet
NULL	NULL

- **SAVEPOINT A;**

QUERY

```
CREATE DATABASE DBMS2_Parinita;
USE DBMS2_Parinita;
SAVEPOINT A;
```

- **INSERT INTO class VALUES (6,’Chris’)**
- **SAVEPOINT B;**

QUERY

```
CREATE DATABASE DBMS2_Parinita;
USE DBMS2_Parinita;
INSERT INTO CLASS VALUES (6,'Chris');
SAVEPOINT B;
SELECT* FROM CLASS;
```

OUTPUT

Result Grid | Filte

ID	Name
1	Bravo
2	Alex
4	Cheng
5	Abhijeet
6	Chris
NULL	NULL

- **INSERT INTO class VALUES (7,’Bravo’);**
- **SAVEPOINT C;**
- **SELECT * FROM class;**
- **ROLLBACK TO B;**
- **SELECT * FROM class;**
- **ROLLBACK TO A;**

QUERY

```
CREATE DATABASE DBMS2_Parinita;
USE DBMS2_Parinita;
INSERT INTO CLASS VALUES (7,'Bravo');
SAVEPOINT C;
SELECT * FROM CLASS;
ROLLBACK TO B;
SELECT * FROM CLASS;
ROLLBACK TO A;
```

OUTPUT

Result Grid | Filter Rows:

ID	Name
1	Bravo
2	Alex
4	Cheng
5	Abhijeet
6	Chris
7	Bravo
NULL	NULL

Part B: Retrieving Records (EMPLOYEES Table)

1. Find out the employee id, names, and salaries of all the employees.

   QUERY

```
CREATE DATABASE DBMS2_Parinita;
USE DBMS2_Parinita;
SELECT Employee_Id, First_Name, Last_Name, Salary
FROM EMPLOYEES;
```

   OUTPUT

Result Grid | Filter Rows: | Exp

Employee_Id	First_Name	Last_Name	Salary
101	John	Austin	5000.00
102	Sara	Smith	4500.00
103	David	Lee	6000.00
104	Anita	Sharma	4800.00
105	Rahul	Verma	5200.00
106	Priya	Kapoor	7000.00
101	John	Austin	5000.00
102	Sara	Smith	4500.00
103	David	Lee	6000.00
104	Anita	Sharma	4800.00
105	Rahul	Verma	5200.00

2. List out the employees who work under manager 100.

   QUERY

```
CREATE DATABASE DBMS2_Parinita;
USE DBMS2_Parinita;
SELECT *
FROM EMPLOYEES
WHERE Manager_Id = 100;
```

   OUTPUT

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Employee_Id	First_Name	Last_Name	Email	Phone_Number	Hire_Date	Job_Id	Salary	Commission_Pct	Manager_Id	Department_Id
101	John	Austin	john@gmail.com	9876543210	2020-01-15	IT_PROG	5000.00	0.10	100	60
103	David	Lee	david@gmail.com	9876543212	2021-06-10	SA_REP	6000.00	0.15	100	80
105	Rahul	Verma	rahul@gmail.com	9876543214	2022-02-12	MK_REP	5200.00	0.12	100	70
101	John	Austin	john@gmail.com	9876543210	2020-01-15	IT_PROG	5000.00	0.10	100	60
103	David	Lee	david@gmail.com	9876543212	2021-06-10	SA_REP	6000.00	0.15	100	80
105	Rahul	Verma	rahul@gmail.com	9876543214	2022-02-12	MK_REP	5200.00	0.12	100	70

3. Find the names of the employees who have a salary greater than or equal to 4800.

QUERY

```
CREATE DATABASE DBMS2_Parinita;
USE DBMS2_Parinita;
SELECT First_Name, Last_Name
FROM EMPLOYEES
WHERE Salary >= 4800;
```

OUTPUT

Result Grid | Filter Rows:

First_Name	Last_Name
John	Austin
David	Lee
Anita	Sharma
Rahul	Verma
Priya	Kapoor
John	Austin
David	Lee
Anita	Sharma
Rahul	Verma

4. List out the employees whose last name is 'AUSTIN'.

QUERY

```
CREATE DATABASE DBMS2_Parinita;
USE DBMS2_Parinita;
SELECT *
FROM EMPLOYEES
WHERE Last_Name = 'Austin';
```

OUTPUT

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Employee_Id	First_Name	Last_Name	Email	Phone_Number	Hire_Date	Job_Id	Salary	Commission_Pct	Manager_Id	Department_Id
101	John	Austin	john@gmail.com	9876543210	2020-01-15	IT_PROG	5000.00	0.10	100	60
101	John	Austin	john@gmail.com	9876543210	2020-01-15	IT_PROG	5000.00	0.10	100	60

5. Find the names of the employees who work in departments 60, 70, and 80.

QUERY

```
CREATE DATABASE DBMS2_Parinita;
USE DBMS2_Parinita;
SELECT First_Name, Last_Name
FROM EMPLOYEES
WHERE Department_Id IN (60,70,80);
```

OUTPUT

Result Grid | Filter Rows:

First_Name	Last_Name
John	Austin
Sara	Smith
David	Lee
Anita	Sharma
Rahul	Verma
Priya	Kapoor
John	Austin
Sara	Smith
David	Lee
Anita	Sharma
Rahul	Verma

6. Display the unique Manager_Id.

QUERY

```
CREATE DATABASE DBMS2_Parinita;
USE DBMS2_Parinita;
SELECT DISTINCT Manager_Id
FROM EMPLOYEES;
```

OUTPUT

Result Grid | Filter

Manager_Id
100
101
102
103

Part C: Updating Records (Client_master Table)

1. Insert five records into the Client_master table.

QUERY

```
USE DBMS2_Parinita;
INSERT INTO Client12 VALUES
('C123','Rohan','MG Road','Delhi','Delhi',110001,6000),
('C124','Amit','Park Street','Kolkata','WB',700001,4000),
('C125','Neha','Brigade Road','Bangalore','Karnataka',560001,8000),
('C126','Priya','FC Road','Pune','Maharashtra',411001,3000),
('C127','Rahul','Anna Salai','Chennai','Tamil Nadu',600001,9000);

SELECT * FROM Client12;
```

OUTPUT

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

ClientNo	Name	Address	City	State	Pincode	Bal_Due
C123	Rohan	MG Road	Delhi	Delhi	110001	6000.00
C124	Amit	Park Street	Kolkata	WB	700001	4000.00
C125	Neha	Brigade Road	Bangalore	Karnataka	560001	8000.00
C126	Priya	FC Road	Pune	Maharashtra	411001	3000.00
C127	Rahul	Anna Salai	Chennai	Tamil Nadu	600001	9000.00

2. Find the names of clients whose bal_due > 5000.

QUERY

```
CREATE DATABASE DBMS2_Parinita;
USE DBMS2_Parinita;
SELECT Name
FROM Client12
WHERE Bal_Due > 5000;
```

OUTPUT

Result Grid

Name
Rohan
Neha
Rahul

3. Change the bal_due of ClientNO "C123" to Rs. 5100.

QUERY

```
USE DBMS2_Parinita;
UPDATE Client12
SET Bal_Due = 5100
WHERE ClientNo = 'C123';

SELECT * FROM Client12;
```

OUTPUT

Result Grid | Filter Rows: | Edit: | Export/Import:

ClientNo	Name	Address	City	State	Pincode	Bal_Due
C123	Rohan	MG Road	Delhi	Delhi	110001	5100.00
C124	Amit	Park Street	Kolkata	WB	700001	4000.00
C125	Neha	Brigade Road	Bangalore	Karnataka	560001	8000.00
C126	Priya	FC Road	Pune	Maharashtra	411001	3000.00
C127	Rahul	Anna Salai	Chennai	Tamil Nadu	600001	9000.00
NULL	NULL	NULL	NULL	NULL	NULL	NULL